

Municipal Services Application

Part 1 & Part 2 — Complete User Guide

AAPD7112 / PROG7312

Windows Forms · .NET Framework 4.8 · C#

Overview

This guide covers everything you need to compile, run, and use the **Municipal Services Application** — a C# .NET Framework 4.8 Windows Forms desktop app that enables South African citizens to report municipal issues and browse local events.

Key Features

- **Report Issues (Part 1)** — log potholes, water leaks, electricity faults, etc.
- **Local Events & Announcements (Part 2)** — search, filter, and get smart recommendations.
- Advanced data structures: SortedDictionary, Dictionary, HashSet, Queue, Stack, PriorityQueue.

Prerequisites

Operating System	Windows 10 or Windows 11
IDE	Visual Studio 2019 or later
.NET Framework	4.8 (download link below if missing)
Windows SDK	Included with Visual Studio workloads

How to Compile

Step 1 — Open the Solution

Launch **Visual Studio**, then:

- Click **File** → **Open** → **Project/Solution**.
- Navigate to the project folder.
- Open **MunicipalServicesApp.sln** (or **MunicipalServicesApp1.csproj**).

Step 2 — Restore / Verify References

The project uses only built-in .NET Framework assemblies — **no NuGet packages are required**. Visual Studio will resolve all references automatically.

Step 3 — Build the Project

Use either method below:

Keyboard shortcut	Ctrl + Shift + B
Menu	Build → Build Solution

Ensure the build completes with **0 errors** before proceeding.

Step 4 — Output Location

After a successful build, the executable is placed at:

```
bin\Debug\MunicipalServicesApp.exe
```

How to Run

Option A — From Visual Studio

With debugger	Press F5
Without debugger	Press Ctrl + F5

Option B — Run the Executable Directly

- Open File Explorer and navigate to the **bin\Debug** folder inside the project.
- Double-click **MunicipalServicesApp.exe**.

Note

.NET Framework 4.8 must be installed on the machine before running the executable directly.

Using the Application

Main Menu

On launch the **Main Menu** is displayed, featuring a sidebar, logo, title, and service cards.

Card / Nav Item	Status
■ Report Issues	Active
■ Local Events	Active
■ Service Status	Disabled — Part 3 placeholder

Navigate using either the **sidebar labels** or the **dashboard cards**.

Report Issues Form (Part 1)

Field	Description
Location	Street address or area of the issue
Category	Dropdown — select the fault type
Description	Detailed description of the problem
Attach Files	Upload supporting images or documents

Progress Bar

Each completed field adds 25% to the engagement progress bar. Contextual guidance messages assist you through the form.

Submitting a Report

- Fill in all four required fields.
- Click **Submit Report**.
- A success message is shown with a unique reference number.
- Choose to submit another report or return to the Main Menu.

Button	Function
Clear Form	Resets all fields to blank
Back	Returns to the Main Menu

Local Events & Announcements (Part 2)

This section demonstrates all required advanced data structures and includes powerful search, filter, and recommendation features.

Search & Filter Options

Filter	Description
Category Filter	All Categories or a specific event category
Date Range	From / To date pickers
Keyword Search	Searches Title, Description, Category, and Location
Search Button	Applies all active filters simultaneously

Recommendation Feature

The application silently tracks your search behaviour. After several searches in the same category, **recommended events** appear ranked by your preferences and event date proximity. Click any recommendation card to view full details.

Advanced Data Structures

Data Structure	Field	Purpose
SortedDictionary<DateTime, List<Event>>	_eventsByDate	Chronological event storage
Dictionary<string, List<Event>>	_eventsByCategory	Fast category lookup
HashSet<string>	_uniqueCategories	Unique category storage
HashSet<DateTime>	_uniqueDates	Unique event dates
Queue<string>	_searchHistory	Last 5 search queries
Stack<Event>	_lastViewedEvents	Last viewed event navigation
PriorityQueue<Event>	Custom impl.	Recommendation ranking

Project Structure

```
MunicipalServicesApp1/  
  ■■■ Models/  
  ■ ■■■ Event.cs    - Event entity (Title, Category, Date, Location,  
Description)  
  ■ ■■■ Issue.cs    - Municipal issue report model  
  ■■■ DataStructures/  
  ■ ■■■ EventRepository.cs - Collections, searching, recommendations  
  ■ ■■■ PriorityQueue.cs   - Custom priority queue  
  ■■■ Forms/  
  ■ ■■■ MainMenuForm.cs    - Dashboard interface  
  ■ ■■■ ReportIssuesForm.cs - Part 1 functionality  
■ ■■■ LocalEventsForm.cs   - Part 2 functionality  
  ■■■ Program.cs           - Application entry point  
  ■■■ MunicipalServicesApp1.csproj
```

Troubleshooting

Problem	Solution
Compilation errors about partial class	Remove auto-generated .Designer.cs files
LocalEventsForm ambiguous constructor	Delete duplicate Designer files
Recommendations not appearing	Perform multiple searches in the same category
Keyword search not finding location	Update EventRepository.cs search logic
Placeholder text missing	Requires Windows 10 + .NET 4.7+
Submit button not working	Ensure all required fields are completed

Part 2 — Marking Criteria Coverage

Requirement	Implementation
Main Menu	Sidebar + dashboard cards — fully functional
Stacks / Queues / Priority Queues	Fully implemented in EventRepository
Dictionaries / Sorted Dictionaries	Implemented in EventRepository
Sets	HashSet displayed in the UI
Recommendation Feature	Smart recommendations via custom PriorityQueue

■ **Tested On**

Windows 11 · Visual Studio 2022 · .NET Framework 4.8 | Part 1 & Part 2 are fully functional with no external dependencies.